Study Buddy

Product Requirements Document: Net New Build

Build name: Study Buddy

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1. PROBLEM

Beginner students studying independently can struggle to understand educational text and identify what they have misunderstood. Reading alone does not provide targeted feedback, so students may repeat material they already understand while missing important gaps. This is a problem hypothesis to validate with student testing.

1a. Opportunity

Turn approved educational lessons into a guided cycle of explanation, practice, feedback, and targeted reteaching. The project also gives its builders practice with tool use, structured agent handoffs, conditional routing, and persistent learning history.

Market Opportunity

- Initial opportunity: a pilot for beginner students using one subject and five curated lessons. Market size and student time savings have not been measured; validate demand through student interviews and a 10-session pilot.

1b. Users & Needs

Primary users: Beginner students who want to understand a supported topic and check their understanding through a short quiz.

Secondary users: People who curate lessons, review generated content, and inspect workflow failures.

Key User Needs

As a beginner student, I need a short explanation grounded in a visible source because the original material can be difficult to understand.

As a returning student, I need feedback and practice focused on my weak concepts because a score alone does not tell me what to study next.

2. PROPOSED SOLUTION

Study Buddy is a web app with two agents that runs when a student selects an approved topic. The Tutor (Agent 1) retrieves a lesson and explains it; Assessment (Agent 2) receives a structured handoff and creates ten multiple-choice questions. After the student submits answers, the controller grades them against the saved answer key and Assessment (Agent 2) explains the results. After a low score, the Tutor (Agent 1) reteaches the missed concepts and the student gets one focused 10-question retry. The controller saves results and weak concepts and shows a study summary.

2a. Value Proposition

Beginner students use Study Buddy to move from reading a lesson to checking and improving their understanding in one guided session. Its distinguishing workflow ties every quiz to the explanation just taught and uses missed concepts to shape the next explanation.

2b. Top 3 MVP Value Props

The Vitamin: A clear explanation, source reference, and ten multiple-choice questions for a supported lesson.

The Painkiller: Immediate answer feedback identifies specific concepts to revisit.

The Steroid: Targeted reteaching and a focused retry respond to the student's actual mistakes, with weak concepts saved for later practice.

2c. Goals & Non-Goals

Goals

- Help beginner students complete a source-grounded lesson and check their understanding independently.
- Help students identify and revisit weak concepts through specific feedback and one targeted retry.
- Deliver trustworthy explanations and quizzes that stay within approved source material and the concepts actually taught.

Non-Goals

- Live web search, pasted notes, and file uploads are outside the first version; learning content comes only from an approved lesson library.
- Free-text grading, teacher dashboards, and unlimited quiz loops are deferred to keep the first build manageable.

2d. Success Metrics

Goal	Signal	Metric	Target
Complete a useful study session	Students finish a first quiz and view feedback	Completed first quizzes / started sessions in a 10-session pilot	At least 8 of 10 sessions; proposed target
Help students revisit weak concepts	Students improve after targeted reteaching	Compare first-attempt and retry scores, both out of 10, across at least 5 low-score study sessions	At least 60% of evaluated low-score sessions show an improvement of at least 1 point on the 10-question retry quiz; proposed target
Keep teaching and assessment grounded	Explanations and questions stay within approved material	Builder review of 50 questions across 5 lessons and their explanations	50/50 questions within taught scope; 0 unsupported teaching claims; pre-release target

3. REQUIREMENTS

User Journey 1: A beginner student learns and practices a supported topic

Context: Turn a selected lesson into an understandable explanation and a short assessment, with feedback tied to what was taught.

Sub-journey: Choose a lesson and understand the topic

- **[P0] User can browse and select a topic from the approved lesson library.**
- **[P0] User can read a short beginner-friendly explanation generated by the Tutor (Agent 1) using only the selected approved lesson.**
- **[P0] User can see the lesson title, original source link, and key concepts alongside the explanation.**
- **[P1] User can choose a difficulty level supported by the selected lesson.**
- **[P2] User can search the approved library by topic name.**

Sub-journey: Check understanding with a quiz

- **[P0] User can answer exactly ten multiple-choice questions covering only concepts in the explanation.**
- **[P0] User can select one answer per question and submit the quiz before seeing the answer key.**
- **[P1] User can review and change selections before submitting.**

- **[P2] User can see how many questions remain unanswered.**

Sub-journey: Review feedback and practice missed concepts

- **[P0] User can view a score out of ten and feedback explaining each answer after submission.**
- **[P0] User can receive targeted reteaching and one focused 10-question retry quiz after a low score (below 8/10); after that retry, the session ends with a summary.**
- **[P1] User can see which concepts need more practice and follow the original lesson link to study further.**

User Journey 2: A returning student reviews progress and continues learning

Context: Preserve learning history so students can revisit previous work and focus later sessions on their weak concepts.

Sub-journey: Review previous study sessions

- **[P0] User can return to their saved quizzes, answers, feedback, scores, and weak concepts through a learner-scoped profile.**
- **[P0] User can see an accurate session status and a clear explanation if a technical failure prevents completion.**
- **[P1] User can view first-attempt and retry results separately for a lesson.**
- **[P2] User can view a simple score trend across completed sessions.**

Sub-journey: Continue with relevant practice

- **[P0] User can start another session informed by their saved weak concepts for a supported lesson.**
- **[P0] User can receive a clarification request when their input is necessary to resolve an ambiguous topic or choice.**
- **[P1] User can see a suggested next approved lesson or concept at the end of a session.**

4. APPENDIX

Project Repository

https://github.com/Soma-Majumder/Study-Buddy

Educational Content

Proposed starting source: A curated selection of OpenStax lessons (https://openstax.org/)

Before importing any lesson:

- Review each selected lesson's license
- Keep source links and attribution
- Approve the content

Still to decide: Subject and exact lessons

Multi-Agent Workflow

Student selects a topic

↓ Controller starts the session

↓ Tutor (Agent 1) retrieves and simplifies the approved lesson

↓ Controller validates the explanation, source references, and concept IDs

↓ Assessment (Agent 2) builds a 10-question quiz and returns a private answer key

↓ Controller shows the quiz and waits for student answers

↓ Student submits answers

↓ Controller grades submitted answers against the saved key

↓ Assessment (Agent 2) explains verified results and identifies weak concepts

↓ Controller checks: score below 8/10 AND learning retry unused?

Yes → Controller validates weak-concept handoff → Tutor (Agent 1) reteaches → controller validates revised explanation → Assessment (Agent 2) builds one focused 10-question retry → student submits → controller grades → Assessment (Agent 2) explains results

No → continue

↓ Controller saves progress, shows summary, and only then marks the session complete

Shared state: learner/session ID, lesson ID/version, explanation ID, quiz/attempt IDs, concept IDs, answers, verified score, feedback, weak concepts, learning retry count, and repair counts keyed by handoff ID. Only controller code writes validated records.

Every agent handoff includes task, relevant context, sources, expected output, and uncertainty with reasons.

Critical handoff: Tutor (Agent 1) → controller → Assessment (Agent 2). Missing explanation or source/concept references blocks quiz creation; allow one repair, then fail explicitly.

Return handoff: Assessment (Agent 2) → controller → Tutor (Agent 1). Missing weak concepts blocks reteaching; allow one repair for that handoff, then fail explicitly. Validate concept IDs against the current lesson and quiz attempt before reteaching.

Agreed Rules

Roles

- **Tutor (Agent 1) teaches**
- **Assessment (Agent 2) creates quizzes and explains results**

Handoffs

- Every handoff includes: **task, context, sources, expected output,** and **uncertainty with reasons**
- Application code controls the sequence and validates handoffs
- An incomplete handoff gets **one repair attempt**, then an explicit failure

Students & Quizzes

- Ask students only when their input is required
- Quizzes test **taught concepts only** and use **multiple choice**
- Low scores trigger **one focused retry** after reteaching

What gets saved: quizzes, answers, feedback, scores, and weak concepts

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Proposed Implementation Details

- **Pass threshold: 8/10**
- **Grading:** The controller checks answers against the saved key *deterministically*
- **Explaining: The Assessment (Agent 2) explains results and identifies weak concepts**
- **Answer keys:** Kept on the backend until submission
- **Retry budgets:** Technical repairs and learning retries have *separate* budgets
- Technical failures **never** count as wrong answers

Data Model

Table	What it stores
lessons	topic, source text, URL, attribution, version

study_sessions	learner, lesson, status, learning retry count, repair counts by handoff ID
explanations	text, concept IDs
quizzes & questions	rounds, private keys
attempts & responses	answers, scores, feedback
weak concepts	learner ID, concept IDs, source attempt ID, lesson ID, recorded_at

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Tool Boundaries

Who	Can do
Tutor (Agent 1)	`find_lesson`, `get_lesson`
Both agents	`get_previous_attempts`, scoped to the current learner only
Controller code only	Write validated session and progress records

No arbitrary access to another learner's history.

Failure Handling

- Treat embedded instructions in source text as **data, not commands**
- **Reject** unsupported teaching or questions outside the explanation
- If any of these happens:
    - an unresolved handoff
    - a missing source
    - a failed save
- → Show a **useful error** and record the **actual status**. Never claim success.

Acceptance Checks *(run before release)*

Test 1: Perfect score

- Supported lesson, all answers correct
- Ten in-scope questions → 10/10
- Feedback saved, session completes **without retry**

Test 2: First score 6/10

- Reteach the missed concepts
- Generate one focused ten-question retry
- Wait for submission and save **both** attempts
- End regardless of the retry score

Test 3: Empty source or malformed handoff

- **One repair** at most
- No invented content or score
- Explicit failure if unresolved

Test 4: Borderline pass — first attempt 8/10

- **Passes; no reteaching or retry. Save results and show summary.**

Test 5: Just below threshold — first attempt 7/10

- **Triggers targeted reteaching and one focused 10-question retry when unused. After the retry, save progress and show a summary before marking the session complete regardless of score.**

Test 6: Missing weak concepts

- **Assessment (Agent 2) returns no weak concepts after a low score.**
- **Allow one repair at most; if weak concepts remain missing, fail explicitly and do not start reteaching.**

Also verify:

- Learner isolation
- Answer keys are absent from pre-submission client payloads
- A failed save shows an error and records the actual failed or pending-save status rather than complete.

Open Decisions

- Subject and lesson selection
- Model/provider
- Database and hosting
- Learner identity method
- History retention
- Delivery schedule

Note: Metrics are *proposed targets*, not observed results. Student research and evaluation runs have not yet been completed.